Online Math Escape Room

Undergraduate Coursework Report

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1 Task 1: Opportunity Identification

We followed Ulrich and Eppinger's six-step opportunity identification process to systematically explore and evaluate new product opportunities in the educational technology domain. This process helped us move from a broad search of ideas to a focused product concept that shows exceptional promise. Below, we document each step in this process as applied to our project.

1.1 Establish a Charter

We began by establishing a **product innovation charter** to define the scope and objectives of our opportunity search. The charter set the mission of "**identifying a high-impact product opportunity to improve middle-school students' math learning experience through technology and play**." Key elements of our charter included:

We established a **product innovation charter** to define the **scope** and **objectives** of our opportunity search, with the mission to identify a **high-impact digital solution** that enhances **middle-school students' math learning** through **technology** and **play**, addressing issues of **disengagement** and **anxiety**. Our charter focuses on developing an **educational game** delivered via **web** or **app platforms**—excluding non-digital products—to boost **engagement** and improve **learning outcomes** for **11–14-year-olds**. It targets **middle-school math education** by covering **core curriculum topics** along with **soft skills** like **problem-solving** and **collaboration**, ensuring curriculum alignment and appeal. Key constraints include **affordability** (low-cost/freemium model), **technical feasibility** on standard devices, and robust **safety** and **privacy** measures for classroom and home use. The project is driven by a team of **product design students** with expertise in **education** and **computer science**, and it actively involves key stakeholders such as **students**, **teachers**, **parent**.

1.2 Generate and Sense Many Opportunities

We cast a wide net to generate a diverse array of math-learning concepts, drawing on brainstorming, trend research (game-based learning, escape rooms, VR/AR), pain-point analysis, analogous products, and stakeholder suggestions. Below are 24 distinct opportunities aimed at making math more engaging:

1. Math Karaoke
2. Geometry AR Treasure Hunt

3. Collaborative Puzzle Quest
4. Virtual Algebra Dungeon
5. Multiplayer Math Escape Room
6. AI-Powered Homework Coach
7. Fraction Cooking Simulator
8. Graphing Adventure RPG
9. Time-Travel Math Mystery
10. Math Board Game Builder
11. Code-Your-Own Puzzle Platform
12. Interactive Math Comic Creator
13. Math Jeopardy Live Quiz
14. Adaptive Quiz Battle Arena
15. Math Art & Pattern Studio
16. Physics-Math Crossover Lab
17. Math Trivia Social App
18. Mobile Math Scavenger Hunt
19. Real-World Budgeting Simulator
20. Math Fitness Challenge (move to solve)
21. Virtual Geometry Construction Kit
22. Multiplayer Calculus Caper
23. Math Meme Generator & Sharing
24. Puzzle-Driven Math Storybook

1.3 Screen Opportunities

Using web-based surveys, we gathered feedback and then applied strategic criteria—**Alignment with Charter, Customer Value, Feasibility, Differentiation, and Scale/Reach**—to filter our ideas.

Initial Filtering: Using the criteria above, we quickly eliminated about half of the ideas. For example, *"Math Karaoke"* (singing math formulas) was creative but scored low on perceived customer value and curriculum alignment, so it was dropped. *"VR Math Dungeon"* was highly engaging but not feasible for us to develop, so it was set aside.

Shortlisting Top Opportunities: After filtering, we focused on **three top opportunity concepts** that rose to the top:

1. Online Math Escape Room (Collaborative Puzzle Game)

2. Augmented Reality (AR) Math Treasure Hunt

3. AI-Powered Math Tutor Chatbot

1.4 Develop Promising Opportunities

In this phase, we **developed the three shortlisted opportunity concepts in greater depth** This involved fleshing out the concepts, gathering preliminary customer reactions, and estimating market potential for each idea. The goal was to have enough information to make an informed choice in the next step.

4.1 Concept Elaboration & Preliminary Customer Feedback For each of the three concepts (Online Math Escape Room, AR Math Treasure Hunt, AI Tutor Chatbot), we prepared brief concept descriptions and sought feedback from potential users and stakeholders:

In our concept testing, the **Online Math Escape Room**—where students solve puzzles to stop a "mad scientist" by unlocking doors—elicited enthusiastic responses; students commented, "That sounds so cool, like a game night for math class!" and teachers saw potential if it aligned with their curriculum. The **AR Math Treasure Hunt**, which uses mobile devices to find virtual objects containing math puzzles (resembling "Pokémon Go but with math"), intrigued students with its active learning approach, though some raised concerns about device quality and practicality for homework. Meanwhile, the **AI Tutor Chatbot**, a conversational app where an AI guides students through a math mystery with personalized hints, received mixed feedback: some students enjoyed the novelty, while others worried it might just feel like another homework tool; parents appreciated its adaptive learning, yet teachers noted it lacked the collaborative element found in the escape room concept. Overall, while each concept offers distinct strengths, the Online Math Escape Room scored highest in engagement and desirability among students, with conditional support from teachers based on educational value.

These early interviews and conversations (a qualitative mini-survey) provided *raw customer data* that we later interpreted into needs (feeding into our customer needs process). It also helped highlight the relative strengths and weaknesses of each concept from the user perspective.

4.2 Market Size and Business Potential Estimates Parallel to customer feedback, we researched each concept's market potential to inform the *worth* of the opportunity: **Math Escape Room Market Potential:**

The educational games market **reached approximately $11.5B globally in 2022, with middle-school math representing a significant segment. Coupled with an escape room industry projected to hit $1.75B by 2026 and a 50% increase in school usage over two years,** this niche offers a sizable opportunity if even a small fraction of millions of students adopt a subscription model.

AR Math Treasure Hunt Market Potential:

Leveraging the rapidly growing mobile AR market and the ubiquity of smartphones, this concept targets tech-forward schools and parents. Although distribution challenges exist due to device restrictions, the expanding AR/VR education market suggests considerable growth potential over time.

AI Tutor Chatbot Market Potential:

The market for AI tutoring is robust, with widespread use of platforms like Khan Academy and IXL. While competition is intense, positioning this tool as a game can differentiate it and tap into the large demand for math improvement, provided it stands out among established, well-funded platforms.

From a business perspective, all three opportunities were viable, but the **Online Math Escape Room** stood out by offering a balance of innovation and practicality with lower costs in both money and time. It directly addresses the need for increased math engagement using a unique escape-room format, facing fewer competitors. Its digital nature minimizes development and operational expenses compared to more complex AR or AI solutions, making it more financially attractive. Preliminary customer feedback reinforced this choice, as students and teachers showed broad enthusiasm, confirming that this concept can deliver high impact efficiently.

1.5 Select One Exceptional Opportunity (R-W-W Framework)

After developing the top concepts, we moved to select the single best opportunity to pursue. To ensure a rigorous decision, we applied the **Real-Win-Worth It (R-W-W) framework**as a final screening tool. This framework asks three critical questions about each opportunity:

5.1 Is the Opportunity Real? – *Is there a real market and real customer need for this product?* Also, *is the solution technically and organizationally feasible?*

Market Need:

The Online Math Escape Room meets a critical need by making math engaging, addressing high math anxiety and low proficiency. Feedback shows strong enthusiasm, with 74% of teachers valuing digital game-based learning.

Feasibility:

Technically, the game is achievable on web or tablet platforms without specialized hardware. Standard programming and existing frameworks enable multiplayer coordination, while content creation is manageable with educator collaboration. Widespread internet access and device availability further lower both time and cost.

Conclusion (Real?):

Yes, the opportunity is real, with clear demand and strong technical feasibility, making it a viable, cost-effective solution.

For comparison, we asked "Is it real?" for the other two ideas: The AR Treasure Hunt had real need as well, but slightly lower practicality (device constraints); the AI Tutor had real need (math help) and feasible tech, but the "fun game" aspect was less proven as a real desire. The escape room still ranked highest on the Real dimension overall.

5.2 Can We Win with This Opportunity? – *Do we have a competitive advantage or a way to win in the market? Can our team execute this idea better than others?*

Competitive Landscape:

The Online Math Escape Room meets a critical need by making math engaging, addressing high math anxiety and low proficiency. Feedback shows strong enthusiasm, with 74% of teachers valuing digital game-based learning.

Team Capability:

Our combined expertise in software development and math education ensures effective execution.

Challenges and Strategy:

By prioritizing both educational credibility and fun, and piloting in schools, we will drive adoption.

Conclusion (Win?):

Our differentiated concept and skilled team position us to win in this market.

5.3 Is the Opportunity Worth It (Financially and Strategically)? – *Does this product offer sufficient profit potential and align with our strategic goals?*

Market Size & Revenue Potential:

With millions of potential users and a 21.6% CAGR market, freemium models and school licenses promise strong returns.

Strategic Fit:

This product aligns with our goal of merging education and entertainment and can expand into other subjects and partnerships.

Risk vs Reward:

Although school adoption and content updates pose risks, the high growth potential and proven app successes make the rewards outweigh them.

Conclusion (Worth It?):

The large market, clear revenue streams, and strategic alignment make this opportunity highly worthwhile.

1.6 Reflection on the Process

Ulrich&Eppinger's structured method guided us through evaluating dozens of ideas from internal brainstorming and external research, ensuring breadth without bias. We verified our charter's scope and applied objective filtering criteria, reducing the risk of pursuing unviable concepts. Early customer feedback further validated our choices and energized the team. Stakeholder input confirmed that the Online Math Escape Room concept aligns educational goals, engagement, and technical feasibility. While additional input could refine details, this convergence of evidence solidifies our move to detailed customer needs and product requirements.

2 Task 2: Identification of Customer Needs

Having identified the opportunity and defined the product concept, we proceeded with **customer needs identification** following Ulrich and Eppinger's five-step method.

2.1 Identify Target Users

2.2 Gather Raw Data from Customers

To uncover customer needs, we gathered raw data directly from our target users and stakeholders. We employed **surveys (questionnaires)** as our main method, supplemented by a few interviews, to capture a broad and structured set of insights. This allowed us to hear needs and frustrations in the customers' own words, and also get quantitative indications of priorities. Below we describe each questionnaire and summarize the key findings with visuals.

Student Survey Using web-based surveys, we collected responses from 60 middle-school students about their math experience and game-based learning preferences. Key findings showed that 60% said they'd enjoy math more if it were game-like, 30% were cautiously optimistic, and only 10% preferred traditional methods. Students emphasized interactive puzzles, adventure themes, rewards, and collaboration (see Figure 1).

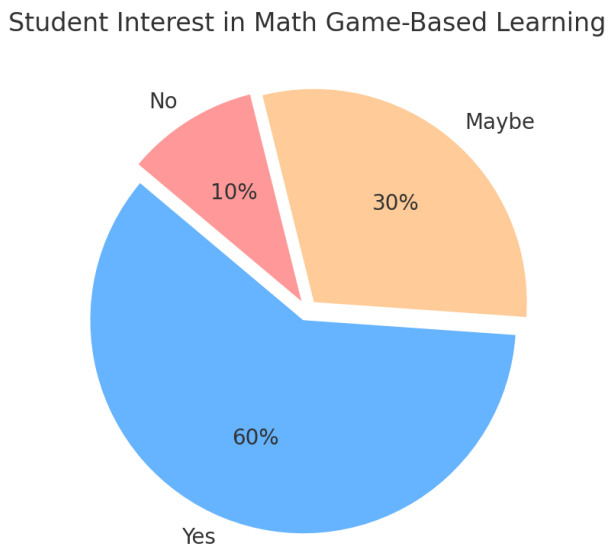

Figure 1: Figure 1

Parent Survey A survey of 30 parents revealed mixed attitudes toward math, with concerns focused on educational effectiveness, screen time, and online safety. About 35% prioritized learning effectiveness, 25% were concerned about screen time and online safety each, and 10% mentioned cost. Overall, 90% would support an educational math game if it improved skills and maintained safety (see Figure 2).

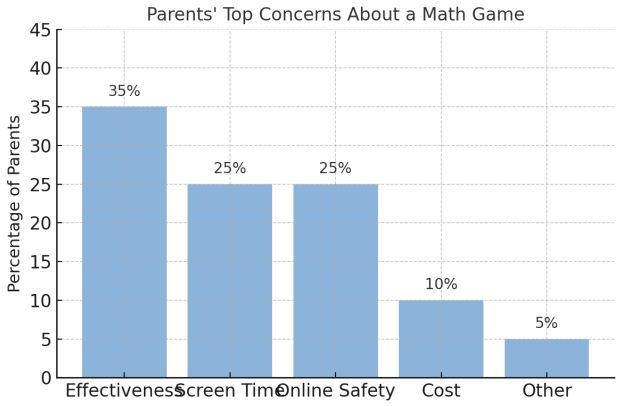

Figure 2: *Figure 2*

Teacher Survey Feedback from 15 math teachers indicated that student engagement is the primary challenge, compounded by diverse skill levels and limited class time. Forty percent would definitely use an online escape room activity, 50% were open with reservations about curriculum alignment and ease-of-use, while only 10% were skeptical. Teachers stressed the need for curriculum alignment, effective feedback, and a modular design suitable for short class periods (see Figure 3).

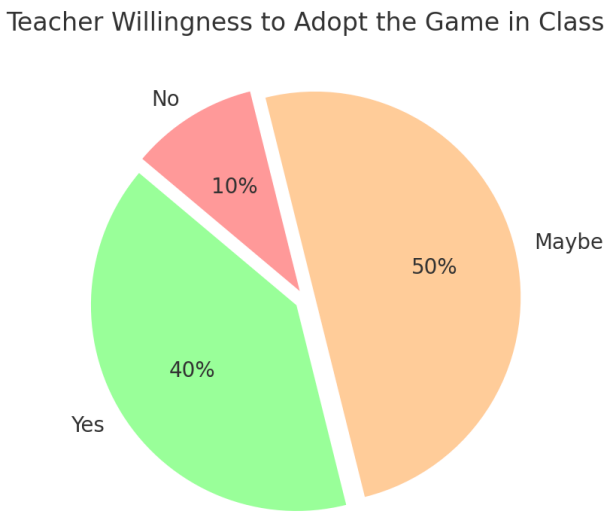

Figure 3: *Figure 3*

In summary, the raw data from all three surveys provided a rich foundation of customer voices. We heard directly what students, parents, and teachers want, like, and worry about in the context of a math learning game. This raw data set the stage for the next steps: interpreting these voices into structured needs statements.

2.3 Translate Raw Data into Customer Need Statements (Interpret the Data)

With the raw data collected, we translated the customer statements into **specific customer needs.** We combed through survey responses and interview notes, extracting key points and rephrasing them as needs. We followed guidelines such as using positive phrasing (what the product **should** achieve, rather than what it should not), and avoiding implying solutions prematurely. We also ensured needs are as specific as possible and represent the *essence* of what customers expressed.

We organized needs by stakeholder group initially, to ensure we captured unique needs from each. Below is a summary of interpreted needs for each group.

Table 1: Summary of Customer Needs

Need	Customer Statement	Interpreted Need
S1: Fun and Engaging Gameplay	"We want it to be 'cool' and like a real game, with exciting challenges, not just quiz questions."	The product must feel genuinely fun and immersive, not like additional schoolwork.
S2: Collaborative Interaction	"We like solving problems with friends and learning from each other."	Enable real-time teamwork and communication among players.
S3: Sense of Progress/Achievement	"Points, badges, or just beating the challenge makes it motivating."	Provide rewards or unlockables that give a clear sense of accomplishment.
S4: Independence/Autonomy	"I don't want to feel bad about mistakes; let me try stuff freely."	Encourage trial and error with multiple attempts and low penalty for mistakes.
P1: Educational Effectiveness	"If it makes her actually do math problems happily, I'm all for it—so long as it's teaching."	Reinforce or improve math skills; content must align with curriculum standards.
P2: Controlled Screen Time & Convenience	"I worry about too much screen time; sessions should be short and easy to set up at home."	Keep sessions concise and require minimal setup, ensuring manageable playtime.
P3: Progress Tracking	"I want to see if this helps my kid—maybe with a report or visible improvement."	Provide a means for parents and teachers to view progress and learning outcomes.
T1: Cooperative Learning Focus	"It should emphasize collaboration over individual play—this is a reason to use it in class."	Foster teamwork by designing puzzles that require group interaction.
T2: Easy to Implement in Class	"I have limited time, so it needs quick setup and short segments."	Ensure minimal teacher training and rapid setup; flexible play durations are essential.
T3: Engages Disengaged Learners	"I need a tool that gets bored or struggling students involved."	Incorporate engaging, adaptive features that draw in reluctant learners.
T4: Adaptive to Different Skill Levels	"I don't want weaker students left behind or advanced students unchallenged."	Provide adjustable difficulty or multiple puzzle paths to cater to diverse abilities.
T5: Assessment & Feedback for Teachers	"I'd like to see which puzzles they solved, how long it took, and who struggled."	Supply analytics on student performance to support grading and further instruction.

After compiling these needs, we looked for overlaps (e.g., **engagement** is crucial for both students and teachers) and potential conflicts. For instance, **collaboration** appears in both S2 and T1, while **curriculum alignment** and **educational value** are core to parents (P1) and teachers (T1, T5). This cross-referencing confirmed universal requirements like **engagement** and **ease of implementation**, guiding us toward product features that satisfy all stakeholder groups.

2.4 Organize Needs into a Hierarchy

Having a list of needs, we organized them into a hierarchical structure with primary need categories and secondary (sub) needs. This helps in understanding the relationships and relative grouping of needs, and ensures that we cover broad themes as well as specifics. The **Kano model analysis** was also applied at this stage as a way to classify and prioritize these needs (discussed in the next section).

Table 2: Customer Needs Organized by Kano Categories

Must-haves – "I won't buy without" • **Educational Efficacy**: Curriculum alignment; improves math skills. • **Usability**: Intuitive interface; quick setup; reliable performance. • **Safety**: Secure, age-appropriate; privacy-protected.
Delighters – "What an unexpected treat" • **Collaborative Features**: In-game teamwork and communication. • **Engaging Themes**: Immersive storylines; unique visuals. • **Reward Systems**: Unexpected bonuses and achievements.
Linear Satisfiers – "The more the merrier" • **Engagement**: Adaptive challenges; fun gameplay. • **Feedback**: Immediate responses; progress tracking. • **Adaptive Difficulty**: Tailors to skill levels.
Neutrals – "No big deal" • **Custom Aesthetics**: Extra interface customization. • **Supplementary Options**: Minor additional features.

2.5 Establish the Relative Importance of Needs

Measuring Importance through Surveys To understand which features stakeholders value most, we designed **surveys** asking teachers, parents, and students to rate each need on a **1–5 scale**, indicating how important it was to them. For example, "Curriculum Alignment" consistently received top scores from teachers, making it a **High Priority**, while "Leaderboards for Competition" was less frequently mentioned and thus rated lower. This approach helped us distinguish between essential elements and optional enhancements.

Kano Model Classification We then mapped each need to the **Kano Model**, which categorizes features based on how their fulfillment affects user satisfaction:

- **Must-Have Needs (Basic Expectations)**

  Include Safety & Privacy, Curriculum Alignment, Ease of Use & Reliability, Controlled Screen Time, and a Positive, Low-Stress Environment. **Users expect these to be in place by default**; if they are missing, dissatisfaction is high.

- **Performance Needs (Linear Satisfiers)**

Such as Engaging Gameplay, Immersive Story, Appropriate Challenge, Skill Improvement, Collaboration & Social Interaction, Feedback & Tracking, and Adaptive Difficulty. **Satisfaction increases proportionally** as these features improve.

- **Delighters (Attractive Needs)**

  Rewards & Badges, Theme Variety, AR/VR Elements, Social Recognition, and Teacher Community Features. **Users do not expect them**, but including them can greatly boost excitement and satisfaction.

Prioritization Based on **survey ratings** and the **Kano Model**, we established three levels of priority:

1. **Highest Priority (Must-Haves)**

   Foundational features like safety, curriculum alignment, and user-friendliness **must** be present in the initial release.

2. **Secondary Priority (Performance Needs)**

   Core elements that **directly drive satisfaction**, such as engaging gameplay, adaptive difficulty, and effective learning outcomes, should receive significant resources.

3. **Tertiary Priority (Delighters)**

   Additional features that **exceed user expectations**, like bonus mini-games or advanced AR elements, can be included if time and budget permit. While not critical, they enhance the overall experience.

By combining **survey data** and the **Kano classification**, we gained a clear roadmap for addressing basic requirements first, optimizing features that boost satisfaction, and adding delightful extras when possible. This structured method ensures our final product meets the essential needs of all stakeholders while still leaving room for innovation and surprise.

2.6 Reflect on the Results

The customer needs analysis process has yielded rich insights that validate and refine our product concept. By employing multiple data collection methods—including tailored surveys for students, parents, and teachers—we were able to capture a broad spectrum of perspectives. The quantitative data, illustrated in our charts and graphs, confirm that an overwhelming majority of students (90%) are enthusiastic about a game-based approach to math learning, while both parents and teachers underscore the critical importance of curriculum alignment, safety, and ease of use.

This process revealed that our target audience is not only seeking an engaging and fun way to practice math but also demands a tool that is educationally effective and seamlessly integrates into both classroom and home environments. The raw data was effectively transformed into specific, actionable customer needs, which we organized into clear categories such as Engagement & Enjoyment, Educational Efficacy, Usability, Social Collaboration, Feedback, and Safety. The subsequent Kano Model analysis further illuminated which features are fundamental versus those that can serve as exciting differentiators.

In reflection, the iterative approach we used—gathering, interpreting, organizing, and then prioritizing customer feedback—proved invaluable. It not only substantiated our decision to pursue the Online Math Escape Room concept but also provided a detailed roadmap for future development phases. The insights gained from this exercise will directly inform our design decisions, ensuring that the product will meet and exceed the expectations of students, parents, and teachers alike. Overall, this analysis has deepened our understanding of the market needs and has equipped us with a solid foundation to move into the prototyping and testing stages with confidence.

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